Speech and Music Presence Detection Across Relative Source Levels

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Abstract

When speech and music overlap, their relative levels can affect whether a detector finds both. This project compares two linear support vector machines (SVMs) using 40 handcrafted features with frozen MATLAB YAMNet. The test used 450 clips from 50 reserved MUSAN source pairs. Each pair supplied two controls and mixtures at seven speech-to-music ratios from -20 to +20 dB. Giving the SVMs equal class priors raised balanced accuracy from 50.00% to 79.88% for speech and from 88.38% to 93.25% for music. Choosing YAMNet thresholds on validation data improved balanced accuracy by 9.88 percentage points for speech and 7.00 for music. All four paired 95% confidence intervals for these gains were above zero. The balanced SVM and YAMNet with validation thresholds had similar overall music scores. With music 20 dB below speech, the balanced SVM detected it in 26 of 50 mixtures and YAMNet in 13.

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1 Introduction

A podcast intro might have someone speaking over a music bed, then a few seconds of music alone. A single speech-or-music label loses information when both sources are present. Here, each clip gets two decisions. The detector reports whether it found speech and whether it found music.

The speech-to-music ratio (SMR) expresses the ratio of speech power to music power in decibels. At negative SMRs, speech is quieter. At positive SMRs, music is quieter. Changing this ratio lets us see how often each detector finds the quieter source.

How does speech-to-music ratio affect the detection of speech and music separately, and how does a linear SVM using handcrafted audio features compare with frozen MATLAB YAMNet on controlled mixtures?

The comparison uses the same held-out clips to test two changes. The SVMs use either the observed class proportions or equal class priors, while YAMNet uses either a fixed threshold or thresholds chosen on the validation set. Balanced accuracy is the main measure because it gives equal weight to detecting a source and correctly recognising its absence.

2 Literature review

2.1 Acoustic features and temporal summaries

Scheirer and Slaney [1] tested 13 acoustic features and several classifiers for speech and music discrimination. Their classifiers performed better when they combined information over longer segments than when they classified individual frames.

The SVM baseline follows that approach by measuring properties such as signal energy and spectral shape on short frames. The mean of each feature describes its typical value in a clip, and the standard deviation shows how much it changes. Two SVMs use the same 40-feature summary to decide separately whether speech and music are present.

2.2 Pretrained audio recognition

Gemmeke et al. [2] built AudioSet from ten-second YouTube clips labelled for a wide range of audio events. Training on AudioSet allows YAMNet to learn from many kinds of recordings before being applied to the speech and music task here.

YAMNet uses a MobileNet-based architecture with outputs for 521 audio events, including Speech and Music [3]. This project uses those two outputs from the pretrained network. A clip receives one score for each target by averaging its scores across the network's input windows.

The MATLAB version of YAMNet uses softmax [4], which gives all 521 scores a shared total of one. The TensorFlow reference uses independent sigmoid scores [3]. In a mixture, both MATLAB scores can be low even when both sources are present. The project therefore compares a fixed 0.5 threshold with thresholds chosen on validation data.

2.3 Detecting overlapping speech and music

Bhattacharjee, Prasanna and Guha [5] address isolated and overlapping speech/music detection using harmonic-percussive features and multi-task learning. Their approach separates harmonic and percussive information before classification and uses related learning tasks to support detection. The work reports evaluation on both generated mixtures and real recordings. It establishes overlap as a distinct detection problem, where source-specific patterns must be recognised within a combined signal.

Their work treats overlap as part of the detection task. The experiment here varies the level of speech relative to music for each pair of excerpts. That makes it possible to track how each detector responds as one source gets quieter.

2.4 Data and evaluation choices

Snyder, Chen and Povey [6] introduced MUSAN for tasks including voice activity detection and speech/music discrimination. Its separately available source recordings make controlled mixing practical. Using one speech track and one music track gives an explicit label for each included source and allows their power ratio to be set before mixing.

Brodersen et al. [7] discuss balanced accuracy for unequal class sizes and the importance of accounting for grouped observations. Each source pair here gives eight positive examples and one negative example for each target. A detector can earn a high F1 score by marking every clip as positive. Balanced accuracy also gives credit for correctly identifying clips where the target is absent. For the confidence intervals, all nine clips from a pair stay together.

3 Methods

3.1 Sources and partitions

All speech and music recordings come from MUSAN [6]. The experiment divides the source files into training, validation and test sets before extracting any clips. The saved split and seed 530582937 fix the order and pairing. For each file, the code converts the audio to mono, resamples it to 16 kHz, takes the middle two seconds and subtracts the mean. Short or nearly silent excerpts trigger an error, which keeps the selected source list fixed.

The final experiment uses 100 training pairs, 30 validation pairs and 50 test pairs. The test pairs use indices 11–60 from each source type in the saved split. These files are separate from the training, validation and original preliminary test files. Each pair gives seven mixtures and two controls, one with speech alone and one with music alone.

Table 1 Source pairs and examples in the final protocol

Partition	Source pairs	Examples	Positives per target	Negatives per target
Training	100	900	800	100
Validation	30	270	240	30
Test	50	450	400	50

Each mixture counts as a positive example for both targets. A speech-only clip counts as positive for speech and negative for music, and a music-only clip has the reverse labels. The test set contains 350 mixtures and 100 controls, giving 400 positives and 50 negatives for each target.

3.2 Controlled mixtures

The mixture is x = s + am, where s is the speech excerpt, m is the music excerpt and a is the music gain. The equations below give the gain needed to reach the target SMR r from the mean-square powers Ps and Pm.

$$r = 10 \log_{10}\left(\frac{P_s}{a^2 P_m}\right)$$

$$a = \sqrt{\frac{P_s}{P_m 10^{r/10}}}$$

The experiment uses SMRs of -20, -10, -5, 0, +5, +10 and +20 dB. The code measures source power after preprocessing and calculates the achieved ratio before adding the sources. It then scales the mixture to a peak amplitude of 0.99, which preserves the power ratio and avoids clipping. The speech-only and music-only controls use the same peak amplitude.

The same 50 test pairs appear at every SMR. Only their relative levels change.

3.3 Frame features and segment summaries

The code divides each clip into 25 ms frames with a 10 ms hop. At 16 kHz, this gives 400 samples per frame. It measures RMS and zero-crossing rate directly from the samples. For the spectral features, it applies a periodic Hann window and computes the power spectrum using a 512-point FFT.

Table 2 Frame measurements used in the 40-feature summary

Measurement	Definition
RMS	Root mean square amplitude of the unwindowed frame
Zero-crossing rate	Fraction of adjacent samples with different signs
Spectral centroid	Mean frequency weighted by spectral power
Spectral flux	Euclidean distance between successive normalised spectra. Zero for the first frame.
Spectral rolloff	Frequency containing 85% of spectral power
Spectral bandwidth	Power-weighted frequency standard deviation
Spectral flatness	Geometric-to-arithmetic mean ratio of spectral power
MFCCs	Thirteen log-mel cepstral coefficients c0-c12

To compute the mel-frequency cepstral coefficients (MFCCs), the code uses 40 triangular HTK-mel filters from 20 Hz to Nyquist. It floors the filter energies at 10^{-12}, takes their natural logarithms and applies an orthonormal discrete cosine transform. It keeps coefficients c0–c12. The calculation omits pre-emphasis, liftering and delta features. Spectral flatness uses the same floor. For a silent frame, centroid, bandwidth, rolloff and flatness are zero. Flux can still record a change from the previous frame.

Each frame gives seven scalar features and 13 MFCCs. For each clip, the code stores the 20 means followed by the 20 sample standard deviations. It drops any incomplete frame at the end.

3.4 Separate linear SVMs

One binary SVM detects speech and another detects music. The code standardises each feature using its training mean and sample standard deviation, with a scale of one for constant features. It applies the same scaling to both SVM variants and to the validation and test data. The SVM's internal standardisation is disabled.

The unweighted SVMs use the class proportions seen in training. The balanced SVMs give the present and absent classes equal priors, which increases the influence of the fewer negative clips. For each target and weighting rule, the code tests C values of 0.01, 0.1, 1 and 10. It chooses the value with the highest validation balanced accuracy, using the smallest C if scores tie.

Each linear SVM scores the standardised features z using a weight vector w and an intercept b.

$$g(z) = w^T z + b$$

The score is a signed margin, with the decision boundary at zero.

3.5 Frozen YAMNet and threshold selection

The code loads YAMNet with audioPretrainedNetwork and prepares each clip with yamnetPreprocess [4]. It finds the Speech and Music outputs by name and averages each score over the clip. The saved run records a SoftmaxLayer with 521 classes. The experiment uses the pretrained network with its original weights and output layer.

The fixed rule labels a source as present when its mean score reaches 0.5. For the second rule, the code tests thresholds from 0 to 1 in steps of 0.05. It chooses the threshold that gives the highest validation balanced accuracy separately for speech and music. If candidates tie, it uses the one closest to 0.5, then the smaller value. The test uses the thresholds chosen on the validation set.

The scores sum to one in each window and still sum to one after averaging. Both mean scores can reach 0.5 only if the remaining classes account for zero. Lower thresholds can therefore let MATLAB YAMNet report both sources in the same clip.

3.6 Performance measures and uncertainty

A true positive (TP) is a correct detection, and a false negative (FN) is a missed source. A true negative (TN) is a correct rejection, and a false positive (FP) is a false alarm. Balanced accuracy averages recall for present sources and specificity for absent sources.

$$BA = \frac{1}{2}\left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right)$$

F1 summarises precision and recall for the present class.

$$F_1 = \frac{2\,TP}{2\,TP + FP + FN}$$

An always-present reference shows how a rule that accepts every clip scores with eight positives per negative. Overall scores use both the controls and the mixtures. Recall at each SMR uses the 50 mixtures at that ratio. The weaker-source curve leaves out 0 dB because the sources have equal power there.

To estimate 95% confidence intervals, the code draws 2,000 bootstrap samples of the 50 test pairs using seed 530582937. It keeps all nine clips from a pair together and uses the same draws for every system. The interval runs from the 2.5th to the 97.5th percentile of the resampled values. The fitted models and selected thresholds stay fixed during resampling.

The final experiment ran on GitHub Actions using MATLAB R2026b Update 1. The project website links to the code, run instructions and results. A GitHub Actions rerun passed all 16 MATLAB tests and matched all eight saved result tables.

4 Results

4.1 Overall detection

Validation selected C = 0.01 for both speech SVMs. For music, it selected C = 1 for the unweighted SVM and C = 0.1 for the balanced SVM. YAMNet used a selected threshold of 0.05 for both outputs. Table 3 and Figure 1 show how the systems scored on the test set.

Table 3 Test performance across 450 examples. F1 refers to the present class.

System	Speech BA (%)	Speech F1 (%)	Music BA (%)	Music F1 (%)
Always present	50.00	94.12	50.00	94.12
Unweighted SVM	50.00	94.12	88.38	97.11
Balanced SVM	79.88	85.23	93.25	95.73
YAMNet fixed 0.5	68.13	53.21	86.13	83.89
YAMNet selected	78.00	71.79	93.13	92.62

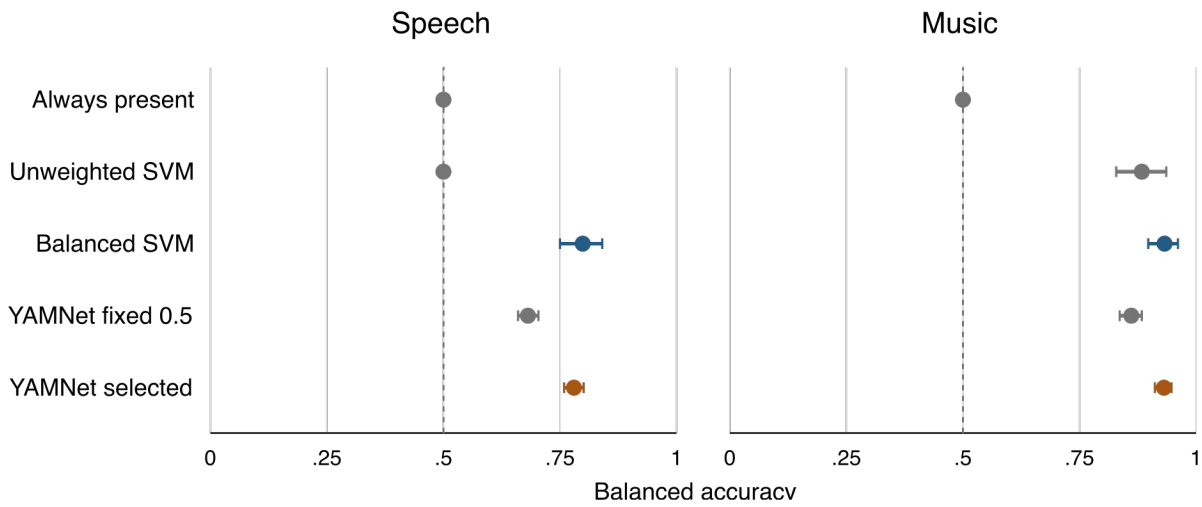


Figure 1 Balanced accuracy and 95% source-pair intervals. The dotted line marks the always-present reference. Models and selected thresholds stay fixed during resampling. Drawn from the recorded MATLAB results.

The balanced SVM reached 79.88% balanced accuracy for speech and 93.25% for music, up 29.88 and 4.88 percentage points from the unweighted SVMs. It correctly rejected 42 of the 50 clips without speech and 47 of the 50 clips without music. The unweighted SVMs rejected zero and 40, respectively.

At 0.05, YAMNet correctly detected 79 more clips with speech and 56 more clips with music than it did at 0.5. Balanced accuracy rose from 68.13% to 78.00% for speech and from 86.13% to 93.13% for music. Both rules had zero false alarms among the 50 negative clips for each target.

The balanced SVM had the highest balanced accuracy for both targets. For music, it scored 93.25%, close to YAMNet's 93.13% at a threshold of 0.05. The unweighted SVMs had the highest F1 among the trained detectors for both targets.

4.2 Paired improvements

Table 4 compares each change with its own baseline using the same test pairs. All four 95% intervals for the change in balanced accuracy lie above zero.

Table 4 Balanced-accuracy gains from 2,000 paired source-pair resamples

Modification	Target	Change (pp)	95% paired interval (pp)
Balanced minus unweighted SVM	Speech	+29.88	[+25.00, +34.06]
Selected minus fixed YAMNet	Speech	+9.88	[+8.13, +11.75]
Balanced minus unweighted SVM	Music	+4.88	[+0.25, +10.13]
Selected minus fixed YAMNet	Music	+7.00	[+5.50, +8.38]

Giving the speech SVM equal class priors produced the largest gain. Its 95% paired interval ran from +25.00 to +34.06 percentage points.

4.3 Detection across relative source levels

Figure 2 shows how often each system detected speech and music at each ratio. Speech recall generally rose as speech got louder relative to music. Music recall generally fell as music got quieter.

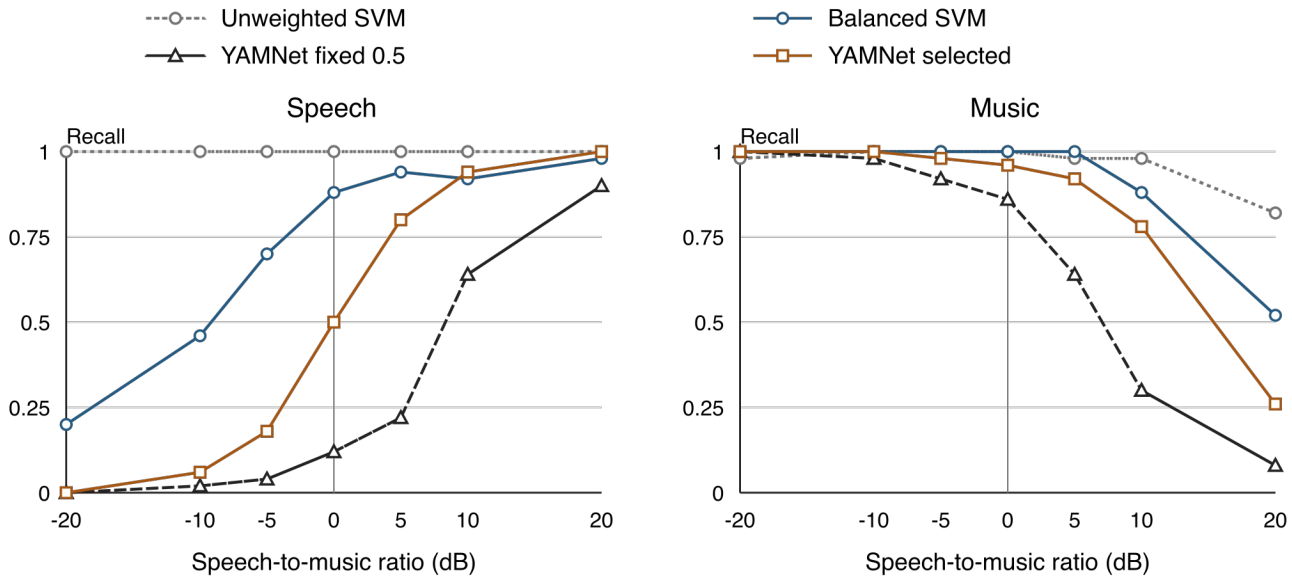


Figure 2 Mixture recall on 50 test pairs at each SMR, drawn from the recorded MATLAB results. Speech is quieter at negative SMRs. Music is quieter at positive SMRs. Specificity is reported separately in Sections 4.1 and 5.1.

When speech was 20 dB quieter than music, the balanced SVM found it in 10 of 50 mixtures, and YAMNet at 0.05 found none. At -5 dB, speech recall was 0.70 for the SVM and 0.18 for YAMNet. At equal power, it reached 0.88 and 0.50. At +10 and +20 dB, YAMNet reached 0.94 and 1.00, compared with 0.92 and 0.98 for the SVM.

The balanced SVM found music in all 50 mixtures at each ratio through +5 dB. Its recall fell to 0.88 at +10 dB and 0.52 at +20 dB. YAMNet at 0.05 reached 0.92, 0.78 and 0.26 at those three positive ratios. At +20 dB, that amounts to 26 detections for the SVM and 13 for YAMNet.

The unweighted speech SVM had perfect recall at every ratio because it labelled every clip as speech. It also labelled all 50 music-only controls as speech, giving it zero specificity.

Of the 350 mixtures, the balanced SVM detected both speech and music in 224 (64%). YAMNet detected both in 119 (34%) at the validation-selected threshold of 0.05 and in none at the fixed threshold of 0.5.

5 Discussion

5.1 How class balance affects detection

The unweighted speech SVM marked every clip as containing speech. That gave it 400 true positives and 50 false positives. Since most clips contained speech, this rule still scored 94.12% on F1. Balanced accuracy was 50.00% because specificity was zero.

Equal class priors give the two classes the same total weight during training. The balanced speech SVM correctly rejected 42 of the 50 music-only clips and detected 303 of the 400 clips with speech. Balanced accuracy rose to 79.88%, and the missed speech clips brought F1 down to 85.23%. If speech detection starts a transcription step, rejecting music-only clips avoids sending them for processing.

For music, false alarms fell from 10 to 3, and missed detections rose from 13 to 30. Specificity increased by 14 percentage points and recall fell by 4.25 points. The average of those two changes gives the 4.88-point gain in balanced accuracy.

5.2 Thresholds connect a pretrained model to the task

At the fixed threshold, YAMNet correctly detected speech in 145 clips and music in 289. With the validation-selected threshold, these counts rose to 224 and 345. Both rules correctly rejected all 50 negative clips for each target.

YAMNet learned to recognise audio events from AudioSet [2, 3]. The MATLAB version uses a softmax output that shares a total score of one across 521 classes [4], so the Speech and Music scores can both be low in a mixture. Validation selected 0.05 for both targets using those softmax scores, with the network weights fixed. The TensorFlow reference uses sigmoid outputs [3].

5.3 What the SMR sweep adds

When music was 20 dB quieter than speech, the balanced SVM detected it in 26 of 50 mixtures, compared with 13 for YAMNet at its selected threshold. This difference is easy to miss in the overall music balanced accuracies of 93.25% and 93.13%.

Speech and music have distinct patterns. The balanced SVM detected more speech at every negative SMR, and YAMNet had higher speech recall at +10 and +20 dB. The balanced SVM had higher music recall at all three positive SMRs.

The louder track may dominate the features or scores of a mixture. Bhattacharjee et al. [5] used harmonic-percussive features to help detect overlapping speech and music. The hardest mixtures in this study, where one source was 20 dB quieter, give a clear condition for testing whether that approach helps.

5.4 Interpreting the comparison

Each source file belonged to one partition. The same test pairs were used at all seven SMRs, so we could examine the effect of changing relative level with the recordings held fixed. The confidence intervals describe uncertainty from resampling these test pairs with the fitted models and selected thresholds fixed.

Labels follow the source tracks. Music with vocals is still labelled as a music source, and a speech excerpt may include a pause. Files by the same speaker or artist may also appear in different partitions. A follow-up could use naturally overlapping recordings labelled by audible events, group files by speaker or artist, and include silence or noise clips to test whether the detectors can reject both targets.

6 Conclusion

The balanced SVM reached 79.88% balanced accuracy for speech and 93.25% for music, gains of 29.88 and 4.88 percentage points over the unweighted SVM. The validation-selected threshold improved MATLAB YAMNet's balanced accuracy by 9.88 points for speech and 7.00 points for music. The four paired improvement intervals were all positive.

The SMR sweep showed where the detectors differed. The balanced SVM detected more speech when speech was quieter. YAMNet detected speech in all 50 mixtures at +20 dB. At that same ratio, the balanced SVM detected music in twice as many mixtures as YAMNet, a difference hidden by their close overall music scores.

7 References

[1] E. Scheirer and M. Slaney, "Construction and evaluation of a robust multifeature speech/music discriminator," in Proc. IEEE ICASSP, vol. 2, pp. 1331-1334, 1997. https://doi.org/10.1109/ICASSP.1997.596192.

[2] J. F. Gemmeke et al., "Audio Set: An ontology and human-labeled dataset for audio events," in Proc. IEEE ICASSP, 2017. Google Research publication.

[3] TensorFlow Models, "YAMNet," model source and documentation. YAMNet repository and output activation configuration. Accessed 13 November 2026.

[4] MathWorks, "audioPretrainedNetwork," Audio Toolbox documentation. MATLAB model documentation. Accessed 13 November 2026.

[5] M. Bhattacharjee, S. R. M. Prasanna and P. Guha, "Clean vs. overlapped speech-music detection using harmonic-percussive features and multi-task learning," IEEE/ACM Trans. Audio, Speech, Lang. Process., vol. 31, pp. 1-10, 2023. https://doi.org/10.1109/TASLP.2022.3164199.

[6] D. Snyder, G. Chen and D. Povey, "MUSAN: A music, speech, and noise corpus," arXiv:1510.08484, 2015. Paper. Corpus and source credits: OpenSLR 17.

[7] K. H. Brodersen et al., "Bayesian mixed-effects inference on classification performance in hierarchical data sets," J. Mach. Learn. Res., vol. 13, no. 101, pp. 3133-3176, 2012. https://www.jmlr.org/papers/v13/brodersen12a.html.